

# Subspaces of Vector Spaces

**Definition 10.1.1** A nonempty subset  $W$  of a vector space  $V$  is called a **subspace** of  $V$  if  $W$  is a vector space under the operations of addition and scalar multiplication defined in  $V$ .

Note that the set  $\{\mathbf{0}\}$  and  $V$  are subspaces of  $V$ .  $\{\mathbf{0}\}$  is called the **zero subspace**. These subspaces are called **trivial subspaces** of  $V$ .

In general, if a nonempty subset  $W$  of  $V$  is given, then one needs to verify all the ten listed axioms for the elements of  $W$  to be a subspace. However, the following theorem reduces this task to only two steps.

**Theorem 10.1.1 (Test for a subspace)**

If  $W$  is a nonempty subset of a vector space  $V$ , then  $W$  is a subspace of  $V$  if and only if the following closure conditions hold.

1. If  $\mathbf{u} \in W$  and  $\mathbf{v} \in W$ , then  $\mathbf{u} + \mathbf{v} \in W$ .
2. If  $\mathbf{u} \in W$  and  $c$  is any scalar, then  $c\mathbf{u} \in W$ .

**Proof.** See Appendix.

**Example 10.1.1.** Show that the set  $W = \left\{ \begin{bmatrix} x \\ 0 \\ z \end{bmatrix} : x, z \in \mathbb{R} \right\}$  is a subspace of  $\mathbb{R}^3$  with the standard operations.

In other words, the  $xz$ -plane in  $\mathbb{R}^3$  is a subspace of  $\mathbb{R}^3$ .

**Solution.** Take  $\mathbf{u} = \begin{bmatrix} x_1 \\ 0 \\ z_1 \end{bmatrix}$  and  $\mathbf{v} = \begin{bmatrix} x_2 \\ 0 \\ z_2 \end{bmatrix}$  from  $W$ . Then

$$\mathbf{u} + \mathbf{v} = \begin{bmatrix} x_1 \\ 0 \\ z_1 \end{bmatrix} + \begin{bmatrix} x_2 \\ 0 \\ z_2 \end{bmatrix} = \begin{bmatrix} x_1 + x_2 \\ 0 \\ z_1 + z_2 \end{bmatrix} \in W.$$

Similarly, if  $c \in \mathbb{R}$ , then

$$c\mathbf{u} = c \begin{bmatrix} x_1 \\ 0 \\ z_1 \end{bmatrix} = \begin{bmatrix} cx_1 \\ 0 \\ cz_1 \end{bmatrix} \in W.$$

## Subspaces of Vector Spaces

---

This shows that  $W$  is closed under addition and scalar multiplication. Therefore, by the Theorem 10.1.1,  $W$  is a subspace of  $\mathbb{R}^3$ . ■

Recall that an  $n \times n$  matrix  $A$  is called *symmetric* if  $A^T = A$ .

For example, if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 0 & -4 \\ 3 & -4 & 5 \end{bmatrix}$$

then  $A^T = A$ . Hence  $A$  is a  $3 \times 3$  symmetric matrix.

But

$$B = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 0 & -4 \\ -3 & 4 & 5 \end{bmatrix}$$

is not a symmetric matrix since  $B^T \neq B$ . (Please verify.)

**Example 10.1.2.** Let  $S_2$  be the set of all  $2 \times 2$  real symmetric matrices. Show that  $S_2$  is a subspace of  $M_{2,2}$ .

**Solution.** Let

$$S_2 = \{A \in M_{2,2} \mid A^T = A\}.$$

We must show that  $S_2$  is closed under addition and scalar multiplication.

Take  $A, B \in S_2$ . Since

$$(A + B)^T = A^T + B^T = A + B.$$

This shows that  $A + B$  is a symmetric matrix. Hence  $A + B \in S_2$ .

Similarly, if  $A \in S_2$  and  $c \in \mathbb{R}$ , then

$$(cA)^T = cA^T = cA.$$

This shows that  $cA$  is also a symmetric matrix. Hence  $cA \in S_2$ . Therefore, by Theorem 10.1.1  $S_2$  is a subspace of  $M_{2,2}$ . In fact, the set of all  $n \times n$  symmetric matrices is a subspace of  $M_{n,n}$ .

## Subspaces of Vector Spaces

---

**Non-Example.** The set of all  $2 \times 2$  singular matrices is NOT a subspace of  $M_{2,2}$ .

Let

$$S = \{A \in M_{2,2} \mid \det A = 0\}.$$

If  $A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ ,  $B = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$ , then clearly,  $A, B \in S$  since  $\det A = 0 = \det B$ .

But

$$A + B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \notin S,$$

since

$$\det(A + B) = \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = 1 \neq 0.$$

**Example 10.1.3.** Let  $D[a, b]$  denote the set of all differentiable real-valued functions defined on the bounded and closed interval  $[a, b]$  with  $a < b$ . Show that  $D[a, b]$  is a subspace of  $C[a, b]$ .

**Solution.** In Example 9.2.4, we proved that  $C[a, b]$ , the set of all continuous functions defined on the bounded and closed interval  $[a, b]$ , is a vector space. From Calculus we know that every differentiable function is continuous. Certainly,  $D[a, b] \subset C[a, b]$ . We want to show that  $D[a, b]$  is subspace of  $C[a, b]$ .

First of all,  $D[a, b] \neq \emptyset$  because the zero function  $0 \in C[a, b]$  defined by  $0(x) = 0$  is differentiable on  $[a, b]$ . Therefore  $0 \in D[a, b]$ .

Take two differentiable functions  $f, g \in D[a, b]$ . Then from Calculus we know that  $f + g$  is also differentiable on  $[a, b]$ . So,  $f + g \in D[a, b]$ .

Similarly, if  $f \in D[a, b]$  then for any scalar  $c \in \mathbb{R}$ ,  $cf$  is also differentiable on  $[a, b]$ . Therefore,

$$cf \in D[a, b].$$

Therefore, by Theorem 10.1.1,  $D[a, b]$  is a subspace of  $C[a, b]$ .

# Subspaces of Vector Spaces

---

## Subspaces of Functions (Calculus)

$W_1 = P[a, b]$  = The set of all polynomial functions that are defined on  $[a, b]$ .

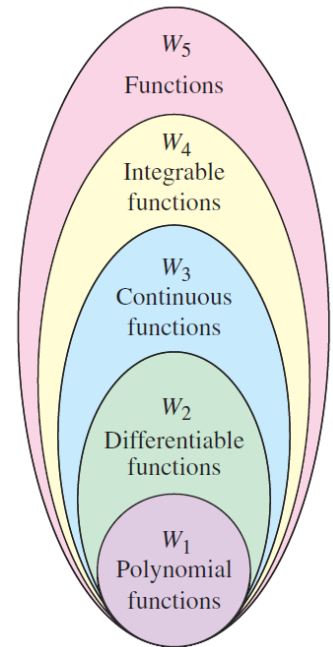
$W_2 = D[a, b]$  = The set of all differentiable functions defined on  $[a, b]$ .

$W_3 = C[a, b]$  = The set of all continuous functions defined on  $[a, b]$ .

$W_4 = R[a, b]$  = The set of all (Riemann) integrable functions defined on  $[a, b]$ .

$W_5 = F[a, b]$  = The set of all functions defined on  $[a, b]$ .

$$W_1 \subset W_2 \subset W_3 \subset W_4 \subset W_5$$



## Theorem 10.1.2. (Intersection of two subspaces is a subspace)

If  $U$  and  $W$  be two subspaces of  $V$ , then the intersection  $U \cap W$  is also a subspace of  $V$ .

**Proof.** See Appendix.